PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

BUMBULIDISTRCT COUNCIL

FORM THREE MIDTERM EXAMINATION

032 **CHEMISTRY**

TIME: 3:00HRS **MARCH 2026**

INSTRUCTIONS:

1. This paper consists of three sections A, Band C with a total of eleven (11) Questions
2. Answer all questions in section A and B, and only two questions from section C
3. All writings should be done in black or blue ink Except drawings which must be in pencil
4. All answers must be written in the space provided for each question.
5. Communication devices and any unauthorized materials are not allowed in the examination room.
6. Write your Name on every page of your answer booklet(s).
7. Whenever necessary use the following atomic masses:

H= 1, Na = 23, N= 14, O= 16, S=32, Ca=40, C= 12

SECTION A(16MARKS)

Answer all questions from this section

1. For items (i) to (x) choose the correct answer from the given alternatives and write its letter in the answer booklet provided

(i) Your friend argued about the scientific procedure that follows after data interpretation. Which step will you suggest to your friend?

A. Observation B. Conclusion C. Hypothesis D. Experimentation

(ii) A large percent of air is composed of:

A. Nitrogen B. Carbon dioxide C. Noble gas D. Oxygen

(iii) Student discovered that it is impossible to light fire in a vacuum due to absence of a certain gas. What comment can you give to a student?

A. Nitrogen is missing C. Oxygen is missing
B. Carbon dioxide is missing D. Hydrogen is missing

(iv) When table salt is dissolved in water, a salt solution is formed. This solution can be separated by a method known as:

A. Sublimation B. Paper chromatography C. Fractional distillation D. Evaporation

(v) The total number of protons and neutrons in the nucleus of an atom is called:

A. Mass number B. Atomic number C. Valence number D Molecular number

(vi) Whichofthefollowingsolutionswillturnpinkwhenphenolphthaleinindicatoris added to it?

A. Orange juice B. Sulphuric acid C. Caustic soda D. Drinking water

(vii) The only substance when dissolved in water to gives out hydrogen ions on its dissociation is simply shown by pH of:

A.7 B.12 C.2 D.10

(viii) Group I element burns in oxygen to form:

A. Non-metal oxide B. Metal oxide C. Carbon dioxide D. Hydroxide

(ix) The oxidation number of phosphorous in the radical PO43-is:

A. +3 B. -5 C. -4 D. +5

(x) Mortar and pestle in the chemistry laboratory is used for

A. Grinding and crushing B. Filtration C. Decantation D. Distillation

2. MatchtheitemsinlistAwiththecorrectresponsesfromlistBbywritingtheletterofthe correct response from list B beside the item number in list A in the space provided.

LIST A	LIST B
(i) A series of investigation (ii) A tentative explanation (iii) A statement that identifies an event facts or situations (iv) A statement of how the results relate to hypothesis (v) A statement in which the researcher explains the results (vi) Used to measure, manipulate and analyze data	A. Observation B. Experimentation C. Hypothesis D. Problem identification E. Data analysis F. Conclusion G. Data collection H. Scientific procedures I. Variables

SECTIONB(54MARKS)

Answer all questions from this section

3. By using a chemical equation, describe any four (4) types of chemical reactions

(b) Study the following chemical reactions

3KOH(aq)+FeCl3(aq)→3KCl(aq)+Fe(OH)(s)

(i) What type of reaction is represented by the chemical equation above?

(ii) Write a net ionic equation for the reaction above

4. The form three students from Mtakuja Secondary school performed an experiment to determine hardness of water by using soap and the following water sample.

Water sample: A = Water from Lembeli

B = Water from Machinjioni

C = Water from Uswahilini

D = Water from Mjini A

SAMPLE	VOLUME OF SOAP USED (cm^3)		Water passed through ion exchanger
	Untreated water	Boiled water	
A	12	1.8	1.8
B	17	17	1.7
C	26	20	1.8
D	1.6	1.6	1.6

(a) Which water sample is hardest? Give a reason for your answer.
(b) Which sample behaves as distilled water? Explain
(c) Name a chemical substance that cause hardness in;
  (i) Water sample A
  (ii) Water sample B
(d) Which problems may people from Lembeli, Machinjioni and Uswahilini face when they use water from their living place?

5. During laboratory experiment, a student used petrol to heat water in a container. The initial mass of the spirit burner with petrol was 120 g and the final mass after burning was 100 g. The heat produced was used to heat 2.5 litres of water. The heat value of petrol is 43640 KJ/kg and the specific heat capacity of water is 4.2 KJ/kg°C
   (a) Determine the mass of petrol that was burnt during the experiment
   (b) Calculate the total heat energy released by the petrol
   (c) Determine the temperature change of the water if all the heat produced was used to heat the water
   (d) State two reasons why the actual temperature rise in a real experiment may be lower than the calculated value

6. (a)A form two student went to the laboratory aimed to perform a certain experiment by using dark grey solid on deflagrating spoon, ignited at high temperature and then the spoon was lowered into the gas jar containing oxygen.
   (i) Identify the dark grey solid that burned
   (ii) Write the balanced chemical equation in the process
   (iii) Explain the nature of the product in terms of acidic or basic properties

   (b) (i) Suppose there is no hydrogen in the universe. What would happen? (two points0
   (ii) Itisfactthatperformanceisdirectlyrelatedtotheaspectsofthatsubstance.Justify the statement based in hydrogen gas. (two points)
7. (a) Some gases are prepared in as special room called fume chamber. How the room support the phenomena? (3 points)
   (b) One of the first aid procedure given to the causality is Resuscitation. As a form three student clarify the process and then deduce any two vitals of the process to the victim.

8. Briefly explain how you can differentiate a solution and suspension by their appearance
   (b) The time taken for sugar to dissolve in cold tea differs from that in hot tea. Explain.
   (c) Explain why cooking oil and water are termed as immiscible liquids

SECTION C (15 Marks)

Answer only one (1) question from this section.

9. The research has revealed that although acids and bases are dangerous to mankind as they are corrosive in nature, yet they are useful to lives of a man. Give four reasons as to why they are useful to man.

10. Dunia Nzima Matatizo was ordered by his father to take the hoes and panga inside the house before going to sleep. A boy forgot the order given by his father. He left them outside and went to sleep. During the morning he found the equipment have been covered by the reddish-brown substance.
    - (a) What is the name given to such reddish substance formed on the surface of the equipment?
    - (b) Write a balanced chemical equation representing the reaction described above?
    - (c) Identify the type of reaction represented by the equation written in (b) above?
    - (d) Explain any four methods which can be used to prevent the chemical process described in (b) above.

11. (a)What do you understand by the following terms as used in chemistry?
    - (i) Empirical formula
    - (ii) Molecular formula
    - (iii) Structural formula

    (b) A certain compound Q is made up of elements Hydrogen(H), Sulphur(S)and Oxygen (O)withthepercentagecompositionof2%,32%andtherestXrespectively.Ifthe molecular mass of compound Q is 98, calculate:
    - (i) The percentage composition X
    - (ii) The molecular formula of compound Q
    - (iii) WritethebalancedionicequationwhenthecompoundQreactwithsodiumhydroxide

    (c) Name the ions found in the following substances
    - (i) Pb(NO3)2
    - (ii) NH4OH

    (d) Using electronic diagram show how CO2compoundis formed

PMO – RALG

BUMBULI DISTRICT COUNCIL

FORM THREE MID-TERM JOINT EXAMINATION

PROPOSED MARKING SCHEME

QN 1

(Award 1 mark each)

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	A	C	D	A	C	C	B	D	A

(Total 10 marks)

QN 2

(Award 1 mark for each correct match)

LIST A	i	ii	iii	iv	v	vi
LIST B	H	C	A	F	E	I

(Total 6 marks)

QN 3

a) Types of Reactions

(Award 1 mark for each correct definition and example. Accept any suitable example.)

i. Combination: A reaction where two or more substances combine to form a single, more complex compound.

Eg. $Mg(s) + O_2(g) \rightarrow MgO(s)$

ii. Decomposition: A reaction where a compound breaks down into two or more simpler substances.

Eg. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

iii. Displacement: A reaction where a more reactive element displaces a less reactive element from its compound.

Eg. $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

iv. Precipitation: A reaction in which two solutions react to form an insoluble solid called a precipitate.

Eg. $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

v. Redox: A reaction that involves both oxidation (loss of electrons/gain of oxygen) and reduction (gain of electrons/loss of oxygen).

Eg. $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(l)$

(Total 5 marks)

b) Reaction Analysis

i. Type of reaction: Double displacement reaction / Precipitation reaction. (1 mark)

ii. Net Ionic Equation:

- Full Ionic: $3K^+(aq) + 3OH^-(aq) + Fe^{3+}(aq) + 3Cl^-(aq) \rightarrow Fe(OH)_3(s) + 3K^+(aq) + 3Cl^-(aq)$
- Net Ionic: $Fe^{3+}(aq) + 3OH^-(aq) \rightarrow Fe(OH)_3(s)$ (3 marks)

(Total 4 marks)

4.(a) (i) Hardest sample: C (uses most soap). (1marks)

(ii) Distilled water: D (same before/after). (1 marks)

(c) Causes: A – Temporary hardness ($Ca(HCO3)2$); B – Permanent hardness ($MgSO4$). (2 marks)

(d) Problems: Scaling, soap wastage, pipe blockage. (3 marks)

QN 5 - Energy and Heat Calculations
a) Mass of petrol burnt: 120g - 100g = 20g = 0.02 kg (2 marks)
b) Heat energy released: Q = heat value × mass = 43640 kJ/kg × 0.02 kg = 872.8 kJ (2 marks)
c) Temperature change of water:
- $Q = mc\Delta T$
- $872.8\ kJ = 2.5\ kg \times 4.2\ kJ/kg°C \times \Delta T$
- $872.8 = 10.5\ \Delta T$
- $\Delta T = 83\ °C$ (3 marks)

d) Reasons for lower actual temperature rise:
(Award 1 mark for any two valid reasons. Total 2 marks)
- Some heat is lost to the surroundings.
- Some heat is absorbed by the container.
- Incomplete combustion of the fuel.
- Heat loss due to evaporation of water.

QN 6
a) Experiment with dark grey solid
i. Identity of solid: The dark grey solid is Silicon (Si). (2 marks)
ii. Balanced equation: $Si(s) + O_2(g) \rightarrow SiO_2(s)$ (2 marks)
iii. Nature of product: The product (Silicon dioxide, SiO_2) is acidic in nature. It will turn moist blue litmus paper red. (2 marks)

b) Importance of Hydrogen
i. If there were no hydrogen in the universe:
(Award 1 mark for each correct point. Total 2 marks)
- Life processes would be affected due to the absence of biological molecules (carbohydrates, proteins, fats) that contain hydrogen.

- Many industrial processes would stop, as hydrogen is a key reactant (e.g., in the Haber process).

ii. Uses of hydrogen related to its properties:

(Award 0.5 marks for any two correct points. Total 1 mark)

- Highly flammable: Used as a fuel and to prepare water gas.
- Acts as a reducing agent: Used to manufacture margarine (hydrogenation of oils).
- Lighter than air: Used to fill weather balloons.

QN 7

a) Significance of a Fume Chamber

(Award 1 mark for each correct point. Total 3 marks)

- It prevents contamination of the laboratory air.
- It protects laboratory users from inhaling harmful/toxic gases.
- It provides a safe environment for conducting experiments that produce hazardous fumes.

b) Resuscitation

- Definition: Resuscitation is the process of restoring breathing and blood circulation to a victim whose breathing or heartbeat has stopped. (1 mark)
- Vitals/Significances of resuscitation:

(Award 1 mark for each correct point. Total 2 marks)

- It restores breathing to the casualty.
- It restores blood circulation, preventing brain damage and death.

QN 8 - Solutions and Suspensions

a) Difference between Solution and Suspension

(Award 1.5 marks for any four correct points of comparison. Total 6 marks)

- A solution is a homogeneous mixture, while a suspension is a heterogeneous mixture.
- A solution is transparent, while a suspension is opaque.
- Particles in a solution dissolve completely and do not settle, while particles in a suspension settle on standing.
- Components of a solution can be separated by evaporation, while particles in a suspension can be separated by filtration.
- A solution does not scatter light, while a suspension shows the Tyndall effect.

b) Dissolving Sugar in Tea

Time taken differs because in cold tea, the molecules move slowly, while in hot tea, the molecules have higher kinetic energy, move faster, and collide more frequently with the sugar, causing it to dissolve faster. (3 marks)

c) Cooking Oil and Water

A mixture of cooking oil and water is an immiscible liquid because they do not mix completely but form two separate layers on standing. (3 marks)

9.Usefulness of acids and bases to man

Introduction (2marks)
Acids and bases are important chemical substances in human life although they are corrosive in nature. Acids are substances that produce hydrogen ions in solution while bases produce hydroxide ions. Despite their dangers, acids and bases have many useful applications in daily activities.

Body (10 marks)

Firstly, acids and bases are used in **agriculture**. Acids such as nitric acid and sulphuric acid are used in the manufacture of fertilizers which improve soil fertility and increase crop yield. Fertilizers are important in ensuring food security and improving living standards. (2.5 marks)

Secondly, acids and bases are used in **food production and preservation**. Acetic acid found in vinegar is used in food preservation and preparation. Citric acid is used in soft drinks and food flavoring to improve taste and quality of food. (2.5 marks)

Thirdly, acids and bases are used in the **medical field**. Some bases such as magnesium hydroxide are used as antacids to neutralize excess stomach acid. Also, some acids are used in the manufacture of drugs used to treat different diseases. (2.5 marks)

Fourthly, acids and bases are used in **industrial processes**. They are used in the manufacture of soap, detergents, plastics, dyes, and other industrial chemicals which are useful in everyday life. (2.5 marks)

Conclusion (3marks)
In conclusion, although acids and bases are corrosive and can be dangerous when mishandled, they are very useful in agriculture, food processing, medicine, and industry. Therefore, proper handling and controlled use of acids and bases should be encouraged

10.

(a) Name: Rust. (1.5 mark)

(b) Process: Rusting/corrosion. (1.5 mark)

(c) Conditions: Water and oxygen. (2 marks)

(d)_ Prevention: Painting, oiling, galvanizing. (2 marks)

QN 11

a) (i). Empirical formula is the one which represent the simplest ratio of the atoms or ions in a compound (2 marks)

(ii). Molecular formula shows the actual number of each atom in a molecule (2 marks)

(iii). Structure formula is a graphic representation of molecular structure showing how the atoms are arranged (2 marks)

b). (i). 65% (1 ½ marks)

(ii). H_2SO_4 (1 ½ marks)

(iii). $2H^+_{(aq)} + 2OH^-_{(aq)} \rightarrow H_2O_{(l)}$ (2 marks)

c). (i). Pb^{+2} and NO^{3-} (2 marks)

d). NH_3^+ and OH^- (2 marks)